

GRAVEATLAS

Phase 12 — Global Preservation, Open Data & Ecosystem Maturity

GRAVEATLAS – PHASE 12
GLOBAL PRESERVATION, OPEN DATA & ECOSYSTEM MATURITY

PHASE 1 THROUGH PHASE 11 MUST ALREADY BE COMPLETE.

Continue from the existing GraveAtlas project.

CORE RULES

- Do NOT rebuild Phases 1-11.
- Do NOT start Phase 13.
- Reuse existing architecture, security, provenance, moderation, audit, APIs, datasets, preservation and release systems.
- No paid services or external integrations without explicit approval.
- Prefer existing/free infrastructure.
- Never expose secrets, credentials, tokens or private data.
- Never fabricate historical facts, records, institutions, users, analytics, coverage, performance or test results.
- Preserve source attribution and provenance.
- Do not silently alter authoritative historical records.
- Do not perform destructive or mass operations without authorization.
- Complete Phase 12 only, then stop.

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PART 1 – PHASE 11 BASELINE

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Read the actual Phase 11 implementation and final report.

Identify:

- completed capabilities
- remaining blockers
- actual limitations
- preservation gaps
- interoperability gaps
- institutional/API maturity
- security/privacy constraints
- operational risks

Use evidence only.

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PART 2 – GLOBAL PRESERVATION NETWORK

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Design a durable preservation model for:

- datasets
- sources
- provenance
- schemas
- releases
- archival snapshots
- documentation

Define preservation responsibilities and verification procedures.

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PART 3 – PRESERVATION HEALTH

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Monitor where practical:

- backup status
- snapshot integrity
- archive completeness
- schema compatibility
- restoration readiness
- provenance integrity

Never claim preservation health without evidence.

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PART 4 – OPEN DATA

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Evaluate public datasets that may be released legally.

For each dataset determine:

- ownership
- licensing
- attribution
- privacy restrictions

- source restrictions
- update frequency
- versioning

Never publish restricted data.

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PART 5 – MACHINE-READABLE DATA
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Where appropriate provide documented formats for:

- cemetery data
- record data
- geographic data
- source metadata
- provenance
- version information

Maintain stable schemas.

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PART 6 – DATASET DOCUMENTATION
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Every public dataset should document where applicable:

- purpose
- scope
- fields
- schema version
- source information
- licensing
- limitations
- update history
- known gaps

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PART 7 – ADVANCED INTEROPERABILITY
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Improve:

- import/export
- schema mapping
- identifier mapping
- migration tooling
- validation

Never silently overwrite authoritative data.

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PART 8 – CROSS-SYSTEM IDENTIFIERS
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Where justified, establish stable identifiers for:

- cemeteries
- records
- sources
- organizations
- datasets
- releases

Identifiers must remain stable across compatible migrations.

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PART 9 – STABLE PUBLIC REFERENCES
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Where appropriate provide stable public references for:

- cemeteries
- records
- collections
- datasets
- sources

Do not expose private records.

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PART 10 – ADVANCED RESEARCH ECOSYSTEM
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Improve researcher capabilities:

- research workspaces
- evidence collections
- citations
- saved searches
- shared collections
- research notes

- source comparison

Protect private research information.

PART 11 – EVIDENCE & PROVENANCE VISUALIZATION

Where practical visualize:

SOURCE

→ RECORD

→ INTERPRETATION

→ REVIEW

→ PUBLICATION

Clearly distinguish:

CONFIRMED

INFERRED

UNCERTAIN

DISPUTED

PART 12 – CHANGE HISTORY

Provide understandable historical change tracking.

Show where appropriate:

- previous version
- new version
- source
- reason
- reviewer
- timestamp

Do not expose private moderation details.

PART 13 – ARCHIVE RESILIENCE

Evaluate:

- backup redundancy
- geographic redundancy
- backup verification
- restoration testing
- disaster recovery
- vendor independence

Do not claim geographic redundancy unless actually implemented.

PART 14 – PORTABILITY

Ensure GraveAtlas can be migrated without losing:

- records
- identifiers
- provenance
- source relationships
- schema information
- documentation

Avoid unnecessary vendor lock-in.

PART 15 – DEVELOPER ECOSYSTEM

Evaluate maturity of:

- public API
- API documentation
- authentication
- authorization
- quotas
- rate limits
- versioning
- monitoring
- abuse prevention

Do not launch features merely because they are technically possible.

PART 16 – API GOVERNANCE

Define:

- acceptable use
- quotas
- authentication requirements
- data restrictions
- version lifecycle
- deprecation policy
- abuse handling

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PART 17 – AI GOVERNANCE

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For AI-assisted research maintain:

- model/version tracking
- input/source references
- generated suggestion
- confidence/uncertainty
- human review
- final decision
- audit history

AI must never silently create authoritative historical facts.

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PART 18 – AI QUALITY CONTROLS

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Evaluate safeguards against:

- hallucinated sources
- invented people
- false relationships
- incorrect OCR
- duplicate false positives
- unsupported historical conclusions

AI output must remain reviewable.

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PART 19 – GLOBAL DATA GOVERNANCE

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Define governance for:

- data stewardship
- source ownership
- institutional contributions
- public releases
- licensing
- corrections
- disputes
- privacy
- API access

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PART 20 – PRIVACY & ETHICS

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Review:

- public/private boundaries
- sensitive information
- research privacy
- organization access
- API exposure
- exports
- AI processing

Apply data minimization and least privilege.

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PART 21 – COMMUNITY MATURITY

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Evaluate:

- contributor quality
- moderation maturity
- dispute resolution
- community research
- institutional participation
- abuse prevention

Do not create unnecessary social features.

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PART 22 – PLATFORM MATURITY

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Score actual maturity in:

- technical reliability
- data quality
- security
- preservation
- interoperability
- community
- institutional readiness
- governance

Use evidence, not assumptions.

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PART 23 – LONG-TERM SUSTAINABILITY

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Evaluate:

- operational continuity
- infrastructure independence
- preservation funding options
- institutional cooperation
- maintenance requirements
- migration strategy

Do not automatically implement monetization.

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PART 24 – DOCUMENTATION

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Create/update as applicable:

docs/GLOBAL-PRESERVATION.md
 docs/PRESERVATION-HEALTH.md
 docs/OPEN-DATA.md
 docs/DATASET-DOCUMENTATION.md
 docs/INTEROPERABILITY.md
 docs/STABLE-IDENTIFIERS.md
 docs/STABLE-REFERENCES.md
 docs/RESEARCH-ECOSYSTEM.md
 docs/PROVENANCE-VISUALIZATION.md
 docs/CHANGE-HISTORY.md
 docs/ARCHIVE-RESILIENCE.md
 docs/PORTABILITY.md
 docs/DEVELOPER-ECOSYSTEM.md
 docs/API-GOVERNANCE.md
 docs/AI-GOVERNANCE.md
 docs/AI-QUALITY.md
 docs/GLOBAL-GOVERNANCE.md
 docs/PLATFORM-MATURITY.md
 docs/LONG-TERM-SUSTAINABILITY.md
 docs/PHASE-12-ROADMAP.md

Document actual implementation only.

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PHASE 12 ACCEPTANCE GATE

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- [] Phase 11 baseline reviewed
- [] Global preservation
- [] Preservation health
- [] Open-data evaluation
- [] Machine-readable datasets
- [] Dataset documentation
- [] Advanced interoperability
- [] Stable identifiers
- [] Stable public references
- [] Research ecosystem
- [] Provenance visualization
- [] Change history
- [] Archive resilience
- [] Portability
- [] Developer ecosystem
- [] API governance
- [] AI governance
- [] AI quality controls
- [] Global data governance
- [] Privacy and ethics
- [] Community maturity
- [] Platform maturity assessment
- [] Long-term sustainability
- [] Documentation

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FINAL REPORT

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PHASE 12 STATUS: READY / NOT READY

For every implemented capability report:

- what was built
- evidence
- tests performed
- limitations
- security considerations
- privacy considerations
- provenance considerations
- operational impact

Use actual metrics only.

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BLOCKERS

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List every issue preventing Phase 12 from being READY.

For each:

- severity
- component
- issue
- evidence
- impact
- remediation
- status

Never include secrets.

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MATURITY SCORECARD

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Technical reliability:
[actual assessment]

Data quality:
[actual assessment]

Security:
[actual assessment]

Preservation:
[actual assessment]

Interoperability:
[actual assessment]

Community:
[actual assessment]

Institutional readiness:
[actual assessment]

Governance:
[actual assessment]

Sustainability:
[actual assessment]

Do not invent numeric scores unless a defined measurement method exists.

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ROADMAP

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NOW:
[items]

NEXT:
[items]

LATER:
[items]

DO NOT BUILD YET:
[items]

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STOP CONDITION

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Do NOT start Phase 13.
Do NOT add paid services without explicit approval.

Do NOT expose secrets.

Do NOT fabricate data or partnerships.

Do NOT silently modify authoritative historical records.

Do NOT publish restricted data.

Do NOT perform destructive bulk operations.

Do NOT perform uncontrolled production load testing.

Do NOT claim preservation, redundancy, recovery or portability capabilities unless they are actually implemented and tested.

STOP after the Phase 12 final report.

WAIT FOR EXPLICIT INSTRUCTIONS FOR PHASE 13.